

Functions of One Real Variable (2022–2026)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS

53

TOTAL MARKS

91 M

TEST DURATION

159 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2022–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	22	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	12	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	19	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q54 • ID: JAM_2026_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54

Let

$$f(x) = \int_0^{e^{\sin x}} \int_0^{\log_e y} e^{-\left(\frac{1}{\sqrt{2}}\right)} (1 - t^2) dt dy.$$

Then $f'(\pi/4) = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 2 (JAM 2026 • Q48 • ID: JAM_2026_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48

$$\int_0^3 (|x - 1| - x[x]) dx = \underline{\hspace{2cm}}$$
 (rounded off to one decimal place).

Question 3 (JAM 2026 • Q44 • ID: JAM_2026_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} + e^{\frac{1 - \cos x}{x}} \right) = \underline{\hspace{2cm}}$$
 (rounded off to one decimal place).

Question 4 (JAM 2026 • Q34 • ID: JAM_2026_Q34)

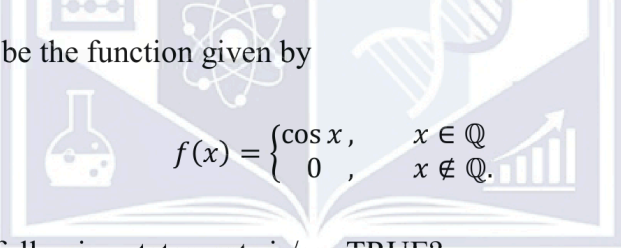
MSQ

+2 M

Neg: Nil (0)

Q.34

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function given by



$$f(x) = \begin{cases} \cos x, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Which of the following statements is/are TRUE?

(A) $f(x)$ is continuous at 0.(B) $f(x)$ is continuous at $\frac{\pi}{2}$.(C) $f(x)$ is Riemann integrable on $[0, 1]$ and $\int_0^1 f(x) dx = \sin 1$.(D) $f(x)$ is Riemann integrable on $[0, 1]$ and $\int_0^1 f(x) dx = 0$.

Question 5 (JAM 2026 • Q33 • ID: JAM_2026_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33	Which of the following functions is/are differentiable at $x = 1$?
(A)	$f(x) = x - 1 ^3$
(B)	$f(x) = x^2 - 1 $
(C)	$f(x) = \begin{cases} x^2 e^{-x^2}, & x \leq 1 \\ e^{-1}, & x > 1 \end{cases}$
(D)	$f(x) = [x]$

Question 6 (JAM 2026 • Q31 • ID: JAM_2026_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.31	Which of the following functions $f: \mathbb{R} \rightarrow \mathbb{R}$ has/have a local minimum at $x = 0$?
(A)	$f(x) = \sin x $
(B)	$f(x) = \sin x + \frac{x^3}{6}$
(C)	$f(x) = x^4 + x^2 + 3$
(D)	$f(x) = \min \{ x - [x], 1 - x + [x] \}$

Question 7 (JAM 2026 • Q21 • ID: JAM_2026_Q21)

MCQ

+2 M

Neg: -0.67

Q.21	Let $f: [1, 9] \rightarrow \mathbb{R}$ be a non-constant continuous function such that $f(1) = f(9)$. Which ONE of the following statements is necessarily TRUE?
(A)	There exists $c \in [4, 8]$ such that $f(c) = f(c + 1)$.
(B)	There exists $c \in [3, 7]$ such that $f(c) = f(c + 2)$.
(C)	There exists $c \in [2, 6]$ such that $f(c) = f(c + 3)$.
(D)	There exists $c \in [1, 5]$ such that $f(c) = f(c + 4)$.

Question 8 (JAM 2026 • Q19 • ID: JAM_2026_Q19)

MCQ

+2 M

Neg: -0.67

Q.19	The value of the following expression $\lim_{x \rightarrow \infty} \int_0^x e^{-t} \sin t dt$ is ____.
(A)	$\frac{e^\pi + 1}{2(e^\pi - 1)}$
(B)	$\frac{e^\pi - 1}{2(e^\pi + 1)}$
(C)	$\frac{2(e^\pi + 1)}{e^\pi - 1}$
(D)	$\frac{2(e^\pi - 1)}{e^\pi + 1}$

Question 9 (JAM 2026 • Q13 • ID: JAM_2026_Q13)

MCQ

+2 M

Neg: -0.67

Q.13	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function that has derivatives of all orders. Which ONE of the following statements is FALSE?
(A)	If f is a polynomial in x of degree at most 5, then $f^{(6)}(x) = 0$ for all $x \in \mathbb{R}$.
(B)	If $f^{(6)}(x) = 0$ for all $x \in \mathbb{R}$, then f is a polynomial in x of degree at most 5.
(C)	If $f^{(k)}(a) = 0$ for all $1 \leq k \leq 5$ and $f^{(6)}(a) > 0$ for some $a \in \mathbb{R}$, then $f(x)$ has a local minimum at a .
(D)	If $f^{(k)}(a) = 0$ for all $1 \leq k \leq 6$ and $f^{(7)}(a) < 0$ for some $a \in \mathbb{R}$, then $f(x)$ has a local maximum at a .

Question 10 (JAM 2026 • Q7 • ID: JAM_2026_Q7)

MCQ

+1 M

Neg: -0.33

Q.7	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(n) = 1$ for all $n \in \mathbb{Z}$. Which ONE of the following statements is necessarily TRUE?
(A)	$f'(n) = 0$ for all $n \in \mathbb{Z}$.
(B)	$f'(x) = 0$ for infinitely many $x \in \mathbb{R} \setminus \mathbb{Z}$.
(C)	$f''(n) = 0$ for all $n \in \mathbb{Z}$.
(D)	$f''(x) = 0$ for infinitely many $x \in \mathbb{R} \setminus \mathbb{Z}$.

Question 11 (JAM 2026 • Q6 • ID: JAM_2026_Q6)

MCQ

+1 M

Neg: -0.33

Q.6	Let $f: (1, 2) \rightarrow \mathbb{R}$ be a continuous function satisfying $\lim_{x \rightarrow 1^+} f(x) = 3 \text{ and } \lim_{x \rightarrow 2^-} f(x) = 3.$ Then which ONE of the following is necessarily TRUE?
(A)	f is bounded and f has a maximum or a minimum but not both.
(B)	f is bounded and f has a maximum or a minimum or both.
(C)	f is bounded and f has neither a maximum nor a minimum.
(D)	f is unbounded and f has either no maximum or no minimum.

Question 12 (JAM 2026 • Q3 • ID: JAM_2026_Q3)

MCQ

+1 M

Neg: -0.33

Q.3	Let $f: (1, 4) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = (f(x) - \pi x)^2 + \pi \quad \text{for all } x \in (1, 4).$ Which ONE of the following is a possible value of $f(3) - f(2)$?
(A)	$\pi + \frac{1}{6}$
(B)	$\pi - \frac{1}{6}$
(C)	$\frac{\pi}{2} + \frac{1}{3}$
(D)	$\frac{\pi}{2} + \frac{1}{2}$

Question 13 (JAM 2025 • Q53 • ID: JAM_2025_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let $f(x) = 2x - \sin x$, for all $x \in \mathbb{R}$. Let $k \in \mathbb{N}$ be such that

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \sum_{i=1}^k i^2 f\left(\frac{x}{i}\right) \right) = 45.$$

Then, the value of k is equal to _____.

Question 14 (JAM 2025 • Q51 • ID: JAM_2025_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying

$$\int_0^{\pi/4} \left(\sin(x) f(x) + \cos(x) \int_0^x f(t) dt \right) dx = \sqrt{2}.$$

Then, the value of $\int_0^{\pi/4} f(x) dx$ is equal to _____ (rounded off to two decimal places).

Question 15 (JAM 2025 • Q43 • ID: JAM_2025_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Let α be the real number such that

$$\lim_{x \rightarrow 0} \frac{(1 - \cos x)(2^{2+x} - 4)}{x^3} = \alpha \ln 2.$$

Then, the value of α is equal to _____ (rounded off to two decimal places).

Question 16 (JAM 2025 • Q42 • ID: JAM_2025_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 The value of

$$\lim_{n \rightarrow \infty} 8n \left(e^{\left(\frac{1}{2n}\right)} - 1 \right) \left(\sin \frac{1}{2n} + \left| \cos \frac{1}{2n} \right| \right)$$

is equal to _____ (rounded off to two decimal places).

Question 17 (JAM 2025 • Q38 • ID: JAM_2025_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that

$$f(0) = 4, f(1) = -2, f(2) = 8 \text{ and } f(3) = 2.$$

Then, which of the following is/are TRUE?

- (A) $|f'(x)| < 5$ for all $x \in [0, 1]$
- (B) $|f'(x_1)| \geq 5$ for some $x_1 \in [0, 1]$
- (C) $f'(x_2) = 0$ for some $x_2 \in [0, 3]$
- (D) $f''(x_3) = 0$ for some $x_3 \in [0, 3]$

Question 18 (JAM 2025 • Q37 • ID: JAM_2025_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that

$$f(0) = 0, f'(0) = 2 \text{ and } f(1) = -3.$$

Then, which of the following is/are TRUE?

- (A) $|f'(x)| \leq 2$ for all $x \in [0, 1]$
- (B) $|f'(x_1)| > 2$ for some $x_1 \in [0, 1]$
- (C) $|f''(x)| < 10$ for all $x \in [0, 1]$
- (D) $|f''(x_2)| \geq 10$ for some $x_2 \in [0, 1]$

Question 19 (JAM 2025 • Q30 • ID: JAM_2025_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 The number of elements in the set

$$\{x \in \mathbb{R} : 8x^2 + x^4 + x^8 = \cos x\}$$

is equal to

(A) 0

(B) 1

(C) 2

(D) greater than or equal to 3

Question 20 (JAM 2025 • Q17 • ID: JAM_2025_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $f(x) = 10x^2 + e^x - \sin(2x) - \cos x$, $x \in \mathbb{R}$. The number of points at which the function f has a local minimum is

(A) 0

(B) 1

(C) 2

(D) greater than or equal to 3

Question 21 (JAM 2025 • Q4 • ID: JAM_2025_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two functions defined by

$$f(x) = \begin{cases} |x|^{1/8} \left| \sin \frac{1}{x} \right| \cos x & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and

$$g(x) = \begin{cases} e^x \cos \frac{1}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}.$$

Then, which one of the following is TRUE?

- (A) f is continuous at $x = 0$, and g is NOT continuous at $x = 0$
- (B) f is NOT continuous at $x = 0$, and g is continuous at $x = 0$
- (C) f is continuous at $x = 0$, and g is continuous at $x = 0$
- (D) f is NOT continuous at $x = 0$, and g is NOT continuous at $x = 0$

Question 22 (JAM 2025 • Q3 • ID: JAM_2025_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be two functions defined by

$$f(x) = \begin{cases} x|x| \left| \sin \frac{1}{x} \right| & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and

$$g(x) = \begin{cases} x^2 \sin \frac{1}{x} + x \cos \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Then, which one of the following is TRUE?

- (A) f is differentiable at $x = 0$, and g is NOT differentiable at $x = 0$
- (B) f is NOT differentiable at $x = 0$, and g is differentiable at $x = 0$
- (C) f is differentiable at $x = 0$, and g is differentiable at $x = 0$
- (D) f is NOT differentiable at $x = 0$, and g is NOT differentiable at $x = 0$

Question 23 (JAM 2024 • Q60 • ID: JAM_2024_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 Define the function $f: (-1, 1) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by

$$f(x) = \sin^{-1} x$$

Let a_6 denote the coefficient of x^6 in the Taylor series of $(f(x))^2$ about

$x = 0$. Then, the value of $9a_6$ equals _____ (rounded off to two decimal places).

Question 24 (JAM 2024 • Q56 • ID: JAM_2024_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3 - 4x^2 + 4x - 6$.

For $c \in \mathbb{R}$, let

$$S(c) = \left\{ x \in \mathbb{R} : f(x) = c \right\}$$

and $|S(c)|$ denote the number of elements in $S(c)$. Then, the value of

$$|S(-7)| + |S(-5)| + |S(3)|$$

equals _____

Question 25 (JAM 2024 • Q52 • ID: JAM_2024_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52 The value of

$$\lim_{t \rightarrow \infty} \left(\left(\log \left(t^2 + \frac{1}{t^2} \right) \right)^{-1} \int_1^{\pi t} \frac{\sin^2 5x}{x} dx \right)$$

equals _____ (rounded off to two decimal places).

Question 26 (JAM 2024 • Q46 • ID: JAM_2024_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Let

$$S = \left\{ f: \mathbb{R} \rightarrow \mathbb{R} : f \text{ is a polynomial and } f(f(x)) = (f(x))^{2024} \text{ for } x \in \mathbb{R} \right\}.$$

Then, the number of elements in S is _____

Question 27 (JAM 2024 • Q38 • ID: JAM_2024_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Define $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ as follows

$$f(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m} (m!)^2} \text{ and } g(x) = \frac{x}{2} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m} (m+1)! m!} \text{ for } x \in \mathbb{R}.$$

Let $x_1, x_2, x_3, x_4 \in \mathbb{R}$ be such that $0 < x_1 < x_2$, $0 < x_3 < x_4$,

$$f(x_1) = f(x_2) = 0, \quad f(x) \neq 0 \text{ when } x_1 < x < x_2,$$

$$g(x_3) = g(x_4) = 0 \quad \text{and } g(x) \neq 0 \text{ when } x_3 < x < x_4.$$

Then, which of the following statements is/are TRUE?

- (A) The function f does not vanish anywhere in the interval (x_3, x_4)
- (B) The function f vanishes exactly once in the interval (x_3, x_4)
- (C) The function g does not vanish anywhere in the interval (x_1, x_2)
- (D) The function g vanishes exactly once in the interval (x_1, x_2)

Question 28 (JAM 2024 • Q34 • ID: JAM_2024_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34 Let $f: (1, \infty) \rightarrow (0, \infty)$ be a continuous function such that for every $n \in \mathbb{N}$, $f(n)$ is the smallest prime factor of n . Then, which of the following options is/are CORRECT?

(A) $\lim_{x \rightarrow \infty} f(x)$ exists

(B) $\lim_{x \rightarrow \infty} f(x)$ does not exist

(C) The set of solutions to the equation $f(x) = 2024$ is finite

(D) The set of solutions to the equation $f(x) = 2024$ is infinite

Question 29 (JAM 2024 • Q29 • ID: JAM_2024_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 Let L_1 denote the line $y = 3x + 2$ and L_2 denote the line $y = 4x + 3$. Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is a four times continuously differentiable function such that the line L_1 intersects the curve $y = f(x)$ at exactly three distinct points and the line L_2 intersects the curve $y = f(x)$ at exactly four distinct points. Then, which one of the following is TRUE?

- (A) $\frac{df}{dx}$ does not attain the value 3 on $\mathbb{R}$
- (B) $\frac{d^2f}{dx^2}$ vanishes at most once on $\mathbb{R}$
- (C) $\frac{d^3f}{dx^3}$ vanishes at least once on $\mathbb{R}$
- (D) $\frac{df}{dx}$ does not attain the value $\frac{7}{2}$ on $\mathbb{R}$

Question 30 (JAM 2024 • Q25 • ID: JAM_2024_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{(x^2 + 1)^2}{x^4 + x^2 + 1} \text{ for } x \in \mathbb{R}.$$

Then, which one of the following is TRUE?

- (A) f has exactly two points of local maxima and exactly three points of local minima
- (B) f has exactly three points of local maxima and exactly two points of local minima
- (C) f has exactly one point of local maximum and exactly two points of local minima
- (D) f has exactly two points of local maxima and exactly one point of local minimum

Question 31 (JAM 2024 • Q24 • ID: JAM_2024_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 Consider the following two statements.

P: There exist functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $g: \mathbb{R} \rightarrow \mathbb{R}$ such that f is continuous at $x = 1$ and g is discontinuous at $x = 1$ but $g \circ f$ is continuous at $x = 1$.

Q: There exist functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $g: \mathbb{R} \rightarrow \mathbb{R}$ such that both f and g are discontinuous at $x = 1$ but $g \circ f$ is continuous at $x = 1$.

Which one of the following holds?

- (A) Both P and Q are true
- (B) Both P and Q are false
- (C) P is true but Q is false
- (D) P is false but Q is true

Question 32 (JAM 2024 • Q19 • ID: JAM_2024_Q19)

MCQ

+2 M

Neg: -0.67

Q.19

For $x \in \mathbb{R}$, let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x .

For $x, y \in \mathbb{R}$, define

$$\min\{x, y\} = \begin{cases} x & \text{if } x \leq y, \\ y & \text{otherwise.} \end{cases}$$

Let $f: [-2\pi, 2\pi] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \sin(\min\{x, x - \lfloor x \rfloor\}) \text{ for } x \in [-2\pi, 2\pi].$$

Consider the set $S = \{x \in [-2\pi, 2\pi]: f \text{ is discontinuous at } x\}$.

Which one of the following statements is TRUE?

- (A) S has 13 elements
- (B) S has 7 elements
- (C) S is an infinite set
- (D) S has 6 elements

Question 33 (JAM 2023 • Q60 • ID: JAM_2023_Q60)

NAT

+2 M

Neg: Nil (0)

Q. 60

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a bijective function such that for all $x \in \mathbb{R}$, $f(x) = \sum_{n=1}^{\infty} a_n x^n$ and $f^{-1}(x) = \sum_{n=1}^{\infty} b_n x^n$, where f^{-1} is the inverse function of f . If $a_1 = 2$ and $a_2 = 4$, then b_1 is equal to _____.

Question 34 (JAM 2023 • Q52 • ID: JAM_2023_Q52)

NAT

+2 M

Neg: Nil (0)

Q. 52 The global minimum value of

$$f(x) = |x - 1| + |x - 2|^2$$

on $\mathbb{R}$ is equal to _____. (rounded off to two decimal places)

Question 35 (JAM 2023 • Q47 • ID: JAM_2023_Q47)

NAT

+1 M

Neg: Nil (0)

Q. 47 Let $f(x) = \sqrt[3]{x}$ for $x \in (0, \infty)$, and $\theta(h)$ be a function such that

$$f(3+h) - f(3) = hf'(3 + \theta(h)h)$$

for all $h \in (-1, 1)$. Then $\lim_{h \rightarrow 0} \theta(h)$ is equal to _____.
(rounded off to two decimal places)

Question 36 (JAM 2023 • Q44 • ID: JAM_2023_Q44)

NAT

+1 M

Neg: Nil (0)

Q. 44 The value of

$$\lim_{n \rightarrow \infty} \left(n \int_0^1 \frac{x^n}{x+1} dx \right)$$

is equal to _____. (rounded off to two decimal places)

Question 37 (JAM 2023 • Q35 • ID: JAM_2023_Q35)

MSQ

+2 M

Neg: Nil (0)

Q. 35 Which of the following functions is/are Riemann integrable on $[0, 1]$?

(A) $f(x) = \int_0^x \left| \frac{1}{2} - t \right| dt$

(B) $f(x) = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

(C) $f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \cap [0, 1] \\ -1 & \text{otherwise} \end{cases}$

(D) $f(x) = \begin{cases} x & \text{if } x \in [0, 1) \\ 0 & \text{if } x = 1 \end{cases}$

Question 38 (JAM 2023 • Q34 • ID: JAM_2023_Q34)

MSQ

+2 M

Neg: Nil (0)

Q. 34 Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function satisfying $f(0) = 0$. Suppose there exists an $M > 0$ such that $|f'(x)| \leq M|x|$ for all $x \in (-1, 1)$. Then

(A) f' is continuous at $x = 0$

(B) f' is differentiable at $x = 0$

(C) ff' is differentiable at $x = 0$

(D) $(f')^2$ is differentiable at $x = 0$

Question 39 (JAM 2023 • Q30 • ID: JAM_2023_Q30)

MCQ

+2 M

Neg: -0.67

Q. 30 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function such that f'' has exactly two distinct zeroes. Then

- (A) f' has at most 3 distinct zeroes
- (B) f' has at least 1 zero
- (C) f has at most 3 distinct zeroes
- (D) f has at least 2 distinct zeroes

Question 40 (JAM 2023 • Q28 • ID: JAM_2023_Q28)

MCQ

+2 M

Neg: -0.67

Q. 28 Let $y : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that y'' is continuous on $[0, 1]$ and $y(0) = y(1) = 0$. Suppose $y''(x) + x^2 < 0$ for all $x \in [0, 1]$. Then

- (A) $y(x) > 0$ for all $x \in (0, 1)$
- (B) $y(x) < 0$ for all $x \in (0, 1)$
- (C) $y(x) = 0$ has exactly one solution in $(0, 1)$
- (D) $y(x) = 0$ has more than one solution in $(0, 1)$

Question 41 (JAM 2023 • Q25 • ID: JAM_2023_Q25)

MCQ

+2 M

Neg: -0.67

Q. 25 Let $f(x) = \cos(x)$ and $g(x) = 1 - \frac{x^2}{2}$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then

- (A) $f(x) \geq g(x)$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- (B) $f(x) \leq g(x)$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- (C) $f(x) - g(x)$ changes sign exactly once on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- (D) $f(x) - g(x)$ changes sign more than once on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Question 42 (JAM 2023 • Q6 • ID: JAM_2023_Q6)

MCQ

+1 M

Neg: -0.33

Q. 6 The limit

$$\lim_{a \rightarrow 0} \left(\frac{\int_0^a \sin(x^2) dx}{\int_0^a (\ln(x+1))^2 dx} \right)$$

is

- (A) 0
- (B) 1
- (C) $\frac{\pi}{e}$
- (D) non-existent

Question 43 (JAM 2023 • Q5 • ID: JAM_2023_Q5)

MCQ

+1 M

Neg: -0.33

Q. 5 Let $p(x) = x^{57} + 3x^{10} - 21x^3 + x^2 + 21$ and

$$q(x) = p(x) + \sum_{j=1}^{57} p^{(j)}(x) \quad \text{for all } x \in \mathbb{R},$$

where $p^{(j)}(x)$ denotes the j^{th} derivative of $p(x)$. Then the function q admits

- (A) NEITHER a global maximum NOR a global minimum on $\mathbb{R}$
- (B) a global maximum but NOT a global minimum on $\mathbb{R}$
- (C) a global minimum but NOT a global maximum on $\mathbb{R}$
- (D) a global minimum and a global maximum on $\mathbb{R}$

Question 44 (JAM 2022 • Q60 • ID: JAM_2022_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60

If $f: [0, \infty) \rightarrow \mathbb{R}$ and $g: [0, \infty) \rightarrow [0, \infty)$ are continuous functions such that

$$\int_0^{x^3+x^2} f(t)dt = x^2 \text{ and } \int_0^{g(x)} t^2 dt = 9(x+1)^3 \text{ for all } x \in [0, \infty),$$

then the value of

$$f(2) + g(2) + 16f(12)$$

is equal to _____. (Rounded off to two decimal places)

Question 45 (JAM 2022 • Q59 • ID: JAM_2022_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59

Let $f: (-1, 1) \rightarrow \mathbb{R}$ and $g: (-1, 1) \rightarrow \mathbb{R}$ be thrice continuously differentiable functions such that $f(x) \neq g(x)$ for every nonzero $x \in (-1, 1)$. Suppose

$$f(0) = \ln 2, \quad f'(0) = \pi, \quad f''(0) = \pi^2, \quad \text{and} \quad f'''(0) = \pi^9$$

and

$$g(0) = \ln 2, \quad g'(0) = \pi, \quad g''(0) = \pi^2, \quad \text{and} \quad g'''(0) = \pi^3.$$

Then the value of the limit

$$\lim_{x \rightarrow 0} \frac{e^{f(x)} - e^{g(x)}}{f(x) - g(x)}$$

is equal to _____. (Rounded off to two decimal places)

Question 46 (JAM 2022 • Q54 • ID: JAM_2022_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54

Let $f: [0, \pi] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} (x - \pi)e^{\sin x} & \text{if } 0 \leq x \leq \frac{\pi}{2}, \\ xe^{\sin x} + \frac{4}{\pi} & \text{if } \frac{\pi}{2} < x \leq \pi. \end{cases}$$

Then the value of

$$\int_0^{\pi} f(x) dx$$

is equal to _____. (Rounded off to two decimal places)

Question 47 (JAM 2022 • Q41 • ID: JAM_2022_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41

The value of the limit

$$\lim_{n \rightarrow \infty} \left(\frac{(1^4 + 2^4 + \dots + n^4)}{n^5} + \frac{1}{\sqrt{n}} \left(\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{4n}} \right) \right)$$

is equal to _____. (Rounded off to two decimal places)

Question 48 (JAM 2022 • Q34 • ID: JAM_2022_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34

Let S be the set of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$|f(x) - f(y)|^2 \leq |x - y|^3 \quad \text{for all } x, y \in \mathbb{R}.$$

Then which of the following is/are true?

(A)

Every function in S is differentiable.

(B)

There exists a function $f \in S$ such that f is differentiable, but f is not twice differentiable.

(C)

There exists a function $f \in S$ such that f is twice differentiable, but f is not thrice differentiable.

(D)

Every function in S is infinitely differentiable.

Question 49 (JAM 2022 • Q33 • ID: JAM_2022_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33

Define $f: [0,1] \rightarrow [0,1]$ by

$$f(x) = \begin{cases} 1 & \text{if } x = 0, \\ \frac{1}{n} & \text{if } x = \frac{m}{n} \text{ for some } m, n \in \mathbb{N} \text{ with } m \leq n \text{ and } \gcd(m, n) = 1, \\ 0 & \text{if } x \in [0,1] \text{ is irrational.} \end{cases}$$

and define $g: [0,1] \rightarrow [0,1]$ by

$$g(x) = \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \in (0,1]. \end{cases}$$

Then which of the following is/are true?

- (A) f is Riemann integrable on $[0,1]$.
- (B) g is Riemann integrable on $[0,1]$.
- (C) The composite function $f \circ g$ is Riemann integrable on $[0,1]$.
- (D) The composite function $g \circ f$ is Riemann integrable on $[0,1]$.

Question 50 (JAM 2022 • Q32 • ID: JAM_2022_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32

Let S be the set of all continuous functions $f: [-1,1] \rightarrow \mathbb{R}$ satisfying the following three conditions:

(i) f is infinitely differentiable on the open interval $(-1,1)$,

(ii) the Taylor series

$$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots$$

of f at 0 converges to $f(x)$ for each $x \in (-1,1)$,

(iii) $f\left(\frac{1}{n}\right) = 0$ for all $n \in \mathbb{N}$.

Then which of the following is/are true?

(A)

$f(0) = 0$ for every $f \in S$.

(B)

$f'\left(\frac{1}{2}\right) = 0$ for every $f \in S$.

(C)

There exists $f \in S$ such that $f'\left(\frac{1}{2}\right) \neq 0$.

(D)

There exists $f \in S$ such that $f(x) \neq 0$ for some $x \in [-1,1]$.

Question 51 (JAM 2022 • Q29 • ID: JAM_2022_Q29)

MCQ

+2 M

Neg: -0.67

Q.29

For $n \in \mathbb{N}$ and $x \in [1, \infty)$, let

$$f_n(x) = \int_0^\pi \left(x^2 + (\cos \theta) \sqrt{x^2 - 1} \right)^n d\theta.$$

Then which one of the following is true?

- (A) $f_n(x)$ is not a polynomial in x if n is odd and $n \geq 3$.
- (B) $f_n(x)$ is not a polynomial in x if n is even and $n \geq 4$.
- (C) $f_n(x)$ is a polynomial in x for all $n \in \mathbb{N}$.
- (D) $f_n(x)$ is not a polynomial in x for any $n \geq 3$.

Question 52 (JAM 2022 • Q27 • ID: JAM_2022_Q27)

MCQ

+2 M

Neg: -0.67

Q.27

Let a, b be positive real numbers such that $a < b$. Given that

$$\lim_{N \rightarrow \infty} \int_0^N e^{-t^2} dt = \frac{\sqrt{\pi}}{2},$$

the value of

$$\lim_{N \rightarrow \infty} \int_0^N \frac{1}{t^2} (e^{-at^2} - e^{-bt^2}) dt$$

is equal to

(A) $\sqrt{\pi}(\sqrt{a} - \sqrt{b}).$

(B) $\sqrt{\pi}(\sqrt{a} + \sqrt{b}).$

(C) $-\sqrt{\pi}(\sqrt{a} + \sqrt{b}).$

(D) $\sqrt{\pi}(\sqrt{b} - \sqrt{a}).$

Question 53 (JAM 2022 • Q22 • ID: JAM_2022_Q22)

MCQ

+2 M

Neg: -0.67

Q.22

Let $H : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $H(x) = \frac{1}{2}(e^x + e^{-x})$ for $x \in \mathbb{R}$.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \int_0^{\pi} H(x \sin \theta) d\theta \quad \text{for } x \in \mathbb{R}.$$

Then which one of the following is true?

- | | |
|-----|--|
| (A) | $xf''(x) + f'(x) + xf(x) = 0$ for all $x \in \mathbb{R}$. |
| (B) | $xf''(x) - f'(x) + xf(x) = 0$ for all $x \in \mathbb{R}$. |
| (C) | $xf''(x) + f'(x) - xf(x) = 0$ for all $x \in \mathbb{R}$. |
| (D) | $xf''(x) - f'(x) - xf(x) = 0$ for all $x \in \mathbb{R}$. |

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of One Real Variable (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	NAT	+2	0.4 to 0.5	19	MCQ	+2	C	37	MSQ	+2	A, B, D
2	NAT	+1	−4.0 to −4.0	20	MCQ	+2	B	38	MSQ	+2	A, C, D
3	NAT	+1	1.0 to 1.0	21	MCQ	+1	A	39	MCQ	+2	A
4	MSQ	+2	B	22	MCQ	+1	A	40	MCQ	+2	A
5	MSQ	+2	A;C	23	NAT	+2	1.50 to 1.70	41	MCQ	+2	A
6	MSQ	+2	A;C;D	24	NAT	+2	5 to 5	42	MCQ	+1	B
7	MCQ	+2	D	25	NAT	+2	0.24 to 0.26	43	MCQ	+1	A
8	MCQ	+2	A	26	NAT	+1	3 to 3	44	NAT	+2	13.39 to 13.41
9	MCQ	+2	D	27	MSQ	+2	B;D	45	NAT	+2	2.00 to 2.00
10	MCQ	+1	B	28	MSQ	+2	B;D	46	NAT	+2	2.00 to 2.00
11	MCQ	+1	B	29	MCQ	+2	C	47	NAT	+1	2.19 to 2.21
12	MCQ	+1	A	30	MCQ	+2	D	48	MSQ	+2	A,D
13	NAT	+2	9 to 9	31	MCQ	+2	A	49	MSQ	+2	A,B,C
14	NAT	+2	1.98 to 2.02	32	MCQ	+2	D	50	MSQ	+2	A,B
15	NAT	+1	1.98 to 2.02	33	NAT	+2	0.49 to 0.51	51	MCQ	+2	C
16	NAT	+1	3.98 to 4.02	34	NAT	+2	0.74 to 0.76	52	MCQ	+2	D
17	MSQ	+2	B;C;D	35	NAT	+1	0.49 to 0.51	53	MCQ	+2	C
18	MSQ	+2	B;D	36	NAT	+1	0.49 to 0.51				

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.